The series $\sum_{n=0}^{\infty} x^n$ converges if $x = \frac{1}{2}$ whereas it does not converge if $x = 2$. This shows that series like $\sum_{n=0}^{\infty} a_n x^n$ converges for some values of x and does not converge for some values of x. The problem of finding the values of x for which sequences of this form converges is beyond the scope of this book. However in the rest of the chapter we list some series with the appropriate values of x for which the series converges and the sum of the series whenever it converges.

5.6.2 Infinite Geometric Series

The series Σx^n is called a geometric series or geometric progression. Let us start with the series: $\sum_{n=0}^{\infty} x^n, x \neq 1$. If $s_n = x_0 + x_1 + x_2 + \cdots + x_n$, then $s_n = \frac{1-x^n}{1-x}$. As x^n tends to 0 if $|x| < 1$, we say that s_n tends to $\frac{1}{1-x}$ if $|x| < 1$.

- $\sum_{n=0}^{\infty} x^n$ converges for all real number x with $|x| < 1$ and the sum is $\frac{1}{1-x}$. That is, for all real numbers x satisfying $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

- $\sum_{n=0}^{\infty} (-1)^n x^n$ converges for all real number x with $|x| < 1$ and the sum is $\frac{1}{1+x}$. That is, for all real numbers x satisfying $|x| < 1$,

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$$

- $\sum_{n=0}^{\infty} (2x)^n$ converges for all real number x with $|x| < \frac{1}{2}$ and the sum is $\frac{1}{1-2x}$. That is, for real numbers x satisfying $|x| < \frac{1}{2}$,

$$\frac{1}{1-2x} = 1 + 2x + 4x^2 + 8x^3 + \cdots$$

- $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges for all real number x and the sum is e^x. That is,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

  for all real numbers x.
- $\sum_{n=0}^{\infty} (-1)^n x$ converges only for $x = 0$.

Let us discuss some special series. By assuming the convergence of those series let us solve some problems.

5.6.3 Infinite Arithmetico-Geometric Series

- The sum of the arithmetico-Geometric series $\Sigma((a + (n-1)d))r^{n-1}$ is given by

$$S = \lim_{n \to \infty} S_n = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

  for $-1 < r < 1$.